MOHEMMED AADIL KHURSHID

AI Application Developer | Rapid Prototyper & Vibe Coding Specialist

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PROFESSIONAL SUMMARY

Results-driven **AI Application Developer** with 4 years of professional experience turning product concepts into high-quality, functional web applications and MVPs at rapid speed. Expert in modern **"vibe coding" workflows** and AI-assisted development using **Cursor, Claude, Gemini, ChatGPT, and v0**. Proven track record at Wayground (Quizizz) architecting end-to-end prototypes, prompt pipelines, multi-model routing, and automated validation. Skilled in ensuring AI-generated code is clean, robust, and aligned with product requirements from initial wireframe to live deployment.

TECHNICAL CORE STACK

Vibe Coding & AI Tools: Cursor IDE, Claude Code, Antigravity, v0 by Vercel, Claude Artifacts, ChatGPT Plus/Canvas, Replit Agent, Git/GitHub.

LLM & Generative AI: Google Gemini (Gemini 3 Flash, 1.5 Pro), Anthropic Claude (3.5 Sonnet, Haiku), OpenAI (GPT-4o, o-series), NaraRouter Gateway, Multi-Model Routing, Prompt Engineering, Multimodal/Vision AI, Structured JSON Extraction.

Web & Prototyping Stack: Next.js (App Router, Turbopack), React, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Vite, Streamlit.

Backend & APIs: Python, FastAPI, Node.js, Express.js, RESTful APIs, Portkey API, Asynchronous Workflows, MongoDB Atlas, PostgreSQL, Docker.

Core Competencies: Rapid MVP Development, Idea-to-Code Translation, Iterative AI Debugging, Code Hygiene & Refactoring, Product Sense, End-to-End Ownership.

PROFESSIONAL EXPERIENCE

WAYGROUND (QUIZIZZ) — **Bangalore, India (Onsite)**

AI Engineer / Consultant, AI/ML Team — *Jan 2026 – Jul 2026*

AI Worksheets — Full-Stack POC for US Market

- Architected and rapidly prototyped a React/Vite frontend and FastAPI backend for US educator workflows, reducing worksheet creation time from hours to minutes using AI-assisted development workflows.
- Engineered production-oriented UI with interactive gameplay matrices, real-time KaTeX mathematical rendering, and clue navigation; structured the frontend as a blueprint for core-site adoption.
- Implemented multi-model routing through Portkey, mapping question tasks to GPT-4o and graphic generation to Gemini 3 Flash to optimize speed and cost efficiency.
- Built graphics backend using headless TikZ/LaTeX subprocess pools and containerized the stack with Docker Compose, PostgreSQL, Redis, and MinIO.

US Mathematics Item Bank Generation Pipeline — Backend

- Built stateful LangGraph generation pipeline with planner, amplification, and question-generation nodes for multi-stage question-bank creation.
- Developed layered automated validation combining schema checks, Vision LLM image verification, and final logical evaluation before upload.
- Designed conditional graphics routing to direct geometry and real-world image requests to SVG/XML processing and abstract mathematical layouts to TikZ.
- Engineered data stratification using Integer Linear Programming and Min/Max Swap logic to match curriculum weightages, DOK metrics, and grade-level targets.

Curriculum-Transformation Engine — Backend ETL

- Built a high-throughput Claude Map-Reduce pipeline that chunks long documents, executes parallel LLM mappings with strict JSON tool calling, and merges outputs from PDFs, DOCX, URLs, and manifests.
- Decoupled schema contracts from parsing logic by deriving layouts and column rules from sample datasets, eliminating hardcoded document scripts.
- Implemented a three-tier integrity gate using Reviewer, Critic, and CSV Audit stages to validate extracted rows before export.

AI Question Generator — Unified Full-Stack Python Application

- Built a portable FastAPI/Streamlit application with local curriculum standards lookup and a GPT-4o verification loop for generated content alignment.

Subject Matter Expert — Mathematics — *Jan 2024 – Dec 2025*

- Authored and curated high-volume K-12 mathematics content aligned with Common Core and TEKS standards.
- Independently architected and shipped a full-stack Next.js web application using OpenAI models to automate math variations and word problems.
- Shipped the application successfully, leading directly to the internal transition into the AI/ML engineering team.

EARLIER EXPERIENCE

- **PERSONAL GOAL PURSUIT** | Career Break (Sep 2022 – Dec 2023) — Structured self-development in quantitative aptitude, logical reasoning, and English.
- **DIGIPPLUS** | Remote | QA/QC Specialist (Jun 2022 – Sep 2022) — Reviewed K-10 mathematics content, structured objective questions, and performed quality audits.
- **INHALTMART & GRAVEIENS EDUSERVICES** | Remote | Subject Matter Specialist (May 2021 – Feb 2022) — Developed mathematics solutions, visual explanations, and structured mathematical content in CMS workflows.

KEY AI PROJECTS & PROTOTYPES

1. VM Copy Question Generator & Multi-LLM Gateway **Next.js 16, TypeScript, AI APIs, Vercel**

- **Rapid Full-Stack MVP:** Built an educational question-generation web app aligned with US curricula standards (Common Core, TEKS) using Next.js 16, React 19, and Tailwind CSS.
- **Multi-Provider AI Gateway:** Engineered a plug-and-play AI client supporting **OpenAI, Google Gemini, and NaraRouter**, enabling real-time model switching (`agnes-3-flash`, `gpt-4o`, `gemini-1.5-pro`) to minimize inference costs.
- **Multimodal Vision Pipeline:** Implemented vision-based document and image analysis to extract tables, graphs, dot plots, and geometric diagrams from uploads and convert them into structured problem sets.
- **Strict Prompt Architecture:** Crafted 3-tier hierarchical prompts enforcing strict JSON array schema outputs, step-by-step verified solutions, and automated distractor rationale.
- **Production Deployment:** Deployed live on Vercel with automated Turbopack builds and secure cloud environment variable management.

2. Engineering Design Automation — AI + Deterministic Systems **FastAPI, React, Python**

- **Concept-to-Execution Prototype:** Built a full-stack AI-assisted engineering design system that converts natural-language requirements into structured technical specifications and SVG drawing artifacts.
- **Hybrid AI Architecture:** Combined LLM natural-language interpretation with deterministic solvers, achieving 424 passing tests across 28 test suites, containerized and deployed with Docker.

3. Atlas AI — Conversational Agent MVP **Personal / Hackathon Project**

- **Idea-to-Product Prototyping:** Built an MVP AI-powered conversational financial assistant with rapid AI-assisted development tools and Telegram bot integration.
- **Stateful Graph:** Implemented query routing, market/news research workflows, and conversational memory extraction using Google Gemini and LangGraph.
- **Validation Framework:** Built the backend with Python and MongoDB Atlas, validating core graph functionality with 59 automated test suites.

4. B2B Textile Marketplace — Rapid Web MVP **Next.js, React, Tailwind CSS**

- **Full-Stack Prototype:** Built a responsive buyer-supplier marketplace using Next.js/React with shared backend architecture and reusable UI components.
- **AI Discovery:** Integrated semantic product discovery and automated buyer-supplier matching workflows.

EDUCATION

B.Tech (Hons.) in Mechanical Engineering **Lovely Professional University | Graduated 2017 (CGPA: 7.66/10)**

Senior Secondary (XII), CBSE — Manas Convent School (Graduated 2013 | 73.20%)

Secondary (X), CBSE — Sunbeam School (Graduated 2011 | CGPA: 9.40/10)

LANGUAGES

English, Hindi, Urdu — Proficient in written and spoken communication.